Earnings Debt Prudence and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Earnings Debt Prudence (EDP), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on EDP achieves an annualized gross (net) Sharpe ratio of 0.39 (0.28), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 21 (15) bps/month with a t-statistic of 2.68 (1.93), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Change in net financial assets, Investment to revenue, Accruals, Sales growth over inventory growth) is 15 bps/month with a t-statistic of 1.98.

1 Introduction

Financial markets efficiently incorporate information about firms' production decisions and investment opportunities into asset prices. The cross-section of stock returns reflects heterogeneous exposure to systematic risks arising from firms' real investment frictions and varying productivity shocks. Despite extensive research documenting how production-based characteristics predict returns, significant gaps remain in understanding how firms' debt management decisions interact with their earnings capacity to affect their cost of capital. We identify and examine a novel predictor of cross-sectional returns: Earnings Debt Prudence (EDP), which captures the relationship between firms' debt financing choices and their earnings generation capacity. This signal provides new insights into how financial constraints and investment frictions manifest in equilibrium asset prices.

In a production-based asset pricing framework, firms face trade-offs between internal and external financing that directly affect their marginal costs of production and investment returns Zhang (2005). Earnings Debt Prudence measures firms' conservative use of debt relative to their earnings capacity, capturing variation in financial flexibility that affects firms' ability to respond to productivity shocks. When firms maintain lower debt levels relative to earnings, they preserve options to invest when positive productivity shocks arrive, reducing their exposure to systematic risk through enhanced operational flexibility Carlson et al. (2004). This operational flexibility becomes particularly valuable during periods of aggregate uncertainty when external financing costs increase and investment opportunities become more dispersed.

The production-based interpretation of EDP relates directly to adjustment costs and investment irreversibility in the cross-section Cooper and Haltiwanger (2006). Firms with high EDP maintain spare debt capacity that reduces the shadow cost of external finance constraints, enabling more efficient capital reallocation when productivity shocks occur. These firms can more readily exercise growth options without facing prohibitive marginal financing costs, leading to lower systematic risk exposure Berk et al. (1999). The ability to time investment optimally in response to productivity signals generates convexity in firm value that manifests as lower required returns in equilibrium.

Furthermore, EDP proxies for heterogeneity in firms' exposure to aggregate investment shocks through the financial accelerator mechanism Bernanke et al. (1999). High EDP firms experience smaller amplification of productivity shocks through the balance sheet channel, as their conservative leverage provides a buffer against tightening financial constraints during economic downturns. This reduced sensitivity to aggregate investment conditions translates into lower conditional risk premia, as these firms' cash flows covary less strongly with the stochastic discount factor Gomes and Schmid (2010). The cross-sectional variation in EDP thus captures differences in how financial frictions interact with real investment decisions to determine equilibrium expected returns.

The value-weighted long/short trading strategy based on EDP achieves an annualized gross Sharpe ratio of 0.39, with monthly average excess returns of 21 basis points (t-statistic = 2.82). The strategy's performance remains robust across multiple factor model specifications, with the monthly alpha relative to the Fama-French six-factor model equaling 21 basis points (t-statistic = 2.68). These results demonstrate economically significant return predictability that persists after controlling for known risk factors. The strategy's gross Sharpe ratio exceeds 81% of documented anomalies in the factor zoo, positioning EDP among the more reliable cross-sectional return predictors.

The EDP signal's predictive power concentrates meaningfully among larger stocks, addressing concerns about small-stock illiquidity and implementation costs. Among stocks with market capitalization above the 80th NYSE percentile, the EDP strategy generates monthly average returns of 24 basis points (t-statistic = 2.75), with six-factor alpha of 20 basis points (t-statistic = 2.25). After accounting for transaction costs using high-frequency bid-ask spreads, the net Sharpe ratio remains economically meaningful at 0.28, exceeding 89% of net Sharpe ratios in the anomaly universe. The strategy's net six-factor alpha equals 15 basis points per month (t-statistic = 1.93), confirming that implementation costs do not eliminate the economic value of the signal.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the EDP strategy maintains a monthly alpha of 15 basis points (t-statistic = 1.98). The signal loads significantly on the HML factor with a coefficient of -0.16 (t-statistic = -4.51) and the CMA factor with a coefficient of 0.19 (t-statistic = 3.64), consistent with the production-based interpretation that high EDP firms maintain investment flexibility. When incorporated into a combination strategy using 155 existing anomalies, adding EDP increases terminal wealth from \$993.77 to \$1,225.43, representing a 23% improvement in cumulative returns over the sample period.

This paper extends the production-based asset pricing literature by documenting how debt prudence relative to earnings capacity captures cross-sectional variation in firms' exposure to investment frictions and productivity shocks. While Whited and Wu (2006) develop financial constraints indices based on structural estimation and Livdan et al. (2009) model the interaction between financial leverage and asset returns through investment irreversibility, our EDP measure provides a simple, directly observable proxy for firms' financial flexibility that strongly predicts returns. Unlike existing leverage-based predictors that focus on debt levels or changes in isolation, EDP explicitly incorporates the earnings dimension, capturing the sustainability of firms' capital structures and their capacity to fund investment from internal resources. This joint consideration of debt and earnings proves crucial for identifying firms with genuine operational flexibility rather than those that are simply under-leveraged relative to industry norms.

Our findings contribute methodologically by demonstrating that EDP's predictive power survives stringent robustness tests following the Novy-Marx and Velikov (2023) protocol, including controls for transaction costs and related anomalies from the factor zoo. The signal's performance among large-cap stocks particularly distinguishes it from many documented anomalies that concentrate in illiquid segments of the market. While Fama and French (2015) argue that profitability and investment factors capture much of the cross-sectional variation in expected returns, our results show that EDP contains incremental information about firms' conditional risk exposures not subsumed by these characteristics. The alpha relative to the six-factor model plus six related anomalies remains statistically significant, suggesting that the interaction between debt capacity and earnings generation represents a distinct dimension of systematic risk.

The broader implications extend to understanding how financial market frictions interact with real investment decisions in determining asset prices. Our evidence supports production-based models where financial constraints affect firms' abilities to respond optimally to productivity shocks, generating cross-sectional variation in risk premia Gomes and Schmid (2010). The documented return predictability of EDP provides empirical support for theoretical frameworks emphasizing the role of financial flexibility in corporate investment and valuation. For practitioners, the signal offers an implementable strategy that generates significant risk-adjusted returns after transaction costs, while for researchers, it highlights the importance of jointly considering multiple dimensions of firms' financial positions when examining cross-sectional return patterns. The results suggest that incorporating measures of debt prudence into asset pricing models could improve our understanding of the sources of systematic risk in financial markets.

2 Data

We obtain financial statement data from the COMPUSTAT annual files, which provide comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over our sample period, which spans several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' debt issuance patterns relative to their operating performance.construction of our Earnings Debt Prudence signal requires two key variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend the firm's long-term obligations. Earnings before interest and taxes (EBIT) measures the firm's operating profitability before considering its capital structure decisions and tax obligations, providing a clean measure of the earnings generated from core business operations. These variables allow us to examine how changes in debt financing relate to the firm's ability to generate operating income.construct the Earnings Debt Prudence signal by calculating the year-over-year change in long-term debt issuance scaled by lagged earnings before interest and taxes. Specifically, we compute (DLTIS(t) - DLTIS(t-1)) / EBIT(t-1) for each firm-year observation. This ratio captures the change in a firm's debt issuance activity relative to its prior-year operating profitability, providing insight into whether firms are accelerating or decelerating their borrowing relative to their earnings capacity. We calculate this measure using fiscal year-end values and require non-missing values for both DLTIS in consecutive years and lagged EBIT. To ensure meaningful ratios, we exclude observations where lagged EBIT equals zero or is negative, as these cases would produce undefined or economically difficult-to-interpret values.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDP signal. Panel A plots the time-series of the mean, median, and interquartile range for EDP. On average, the cross-sectional mean (median) EDP is -0.58 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input EDP data. The signal's interquartile range spans -0.59 to 0.74. Panel B of Figure 1 plots the time-series of the coverage of the EDP signal for the CRSP universe. On average, the EDP signal is available for 6.19% of CRSP names, which on average make up 7.36% of total market capitalization.

4 Does EDP predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDP using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDP portfolio and sells the low EDP portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDP strategy earns an average return of 0.21% per month with a t-statistic of 2.82. The annualized Sharpe ratio of the strategy is 0.39. The alphas range from 0.21% to 0.26% per month and have t-statistics exceeding 2.68 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.16, with a t-statistic of -4.51 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 527 stocks and an average market capitalization of at least $1,196 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 11 bps/month with a t-statistics of 2.54. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for nine exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -15-18bps/month. The lowest return, (-15 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -2.55. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDP trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the EDP strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDP, as well as average returns and alphas for long/short trading EDP strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the EDP strategy achieves an average return of 24 bps/month with a t-statistic of 2.75. Among these large cap stocks, the alphas for the EDP strategy relative to the five most common factor models range from 20 to 28 bps/month with t-statistics between 2.25 and 3.22.

5 How does EDP perform relative to the zoo?

Figure 2 puts the performance of EDP in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EDP strategy falls in the distribution. The EDP strategy's gross (net) Sharpe ratio of 0.39 (0.28) is greater than 81% (89%) of anomaly Sharpe ratios,

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDP strategy (red line).[2] Ignoring trading costs, a \$1 invested in the EDP strategy would have yielded \$2.23 which ranks the EDP strategy in the top 9% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDP strategy would have yielded \$1.23 which ranks the EDP strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDP relative to those. Panel A shows that the EDP strategy gross alphas fall between the 48 and 68 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDP strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDP ranks between the 63 and 80 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does EDP add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDP with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDP or at least to weaken the power EDP has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP} EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

EDP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDP and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDP signal in these Fama-MacBeth regressions exceed 0.20, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on EDP is 0.43.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDP strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDP strategy earns alphas that range from 16-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.08, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDP trading strategy achieves an alpha of 15bps/month with a t-statistic of 1.98.

7 Does EDP add relative to the whole zoo?

Finally, we can ask how much adding EDP to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the EDP signal.[4] We consider

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes EDP grows to \$1225.43.

8 Conclusion

This study provides compelling evidence that Earnings Debt Prudence (EDP) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDP maintains economically and statistically significant predictive power even after accounting for transaction costs and controlling for established risk factors.

The key findings reveal that a value-weighted long/short strategy based on EDP generates substantial risk-adjusted returns, with an annualized gross Sharpe ratio of 0.39 and a net Sharpe ratio of 0.28 after accounting for implementation costs. The strategy's robustness is further evidenced by its ability to deliver significant alpha relative to the Fama-French five-factor model plus momentum, with gross monthly abnormal returns of 21 basis points (t-statistic = 2.68). Notably, even after controlling for six closely related strategies from the factor zoo—including changes in financial liabilities, net debt financing, and various investment and accrual measures—the EDP signal retains a gross monthly alpha of 15 basis points with marginal statistical significance (t-statistic = 1.98).

ization on CRSP in the period for which EDP is available.

These results have important practical implications for investors and portfolio managers. The persistence of the EDP premium after transaction costs suggests that the signal could be profitably implemented in real-world trading strategies, particularly for institutional investors with access to lower transaction costs. The signal's ability to generate alpha beyond existing factors indicates that EDP captures a distinct dimension of firm fundamentals related to prudent debt management that is not fully reflected in current asset prices.

However, several limitations of this study warrant consideration. First, while the net returns remain positive, the reduction in Sharpe ratio from 0.39 to 0.28 and the decline in statistical significance after transaction costs (from t-statistic of 2.68 to 1.93) suggest that implementation frictions materially impact the strategy's profitability. Second, the marginal statistical significance of the alpha relative to the extended factor model including related strategies raises questions about whether EDP represents a truly independent pricing anomaly or partially overlaps with existing factors.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the EDP premium would enhance our understanding of why markets appear to misprice firms based on their earnings-debt relationships. Second, examining the signal's performance across different market conditions, particularly during credit cycles and periods of financial stress, could reveal important time-variation in the strategy's effectiveness. Third, exploring potential enhancements to the EDP signal through machine learning techniques or combination with other fundamental indicators might improve its predictive power. Finally, extending the analysis to international markets would test the generalizability of the findings and potentially uncover cross-country variations in the pricing of debt prudence. Understanding these dimensions would not only advance our theoretical knowledge of asset pricing but also inform more effective investment strategies based on fundamental analysis of corporate financial health.

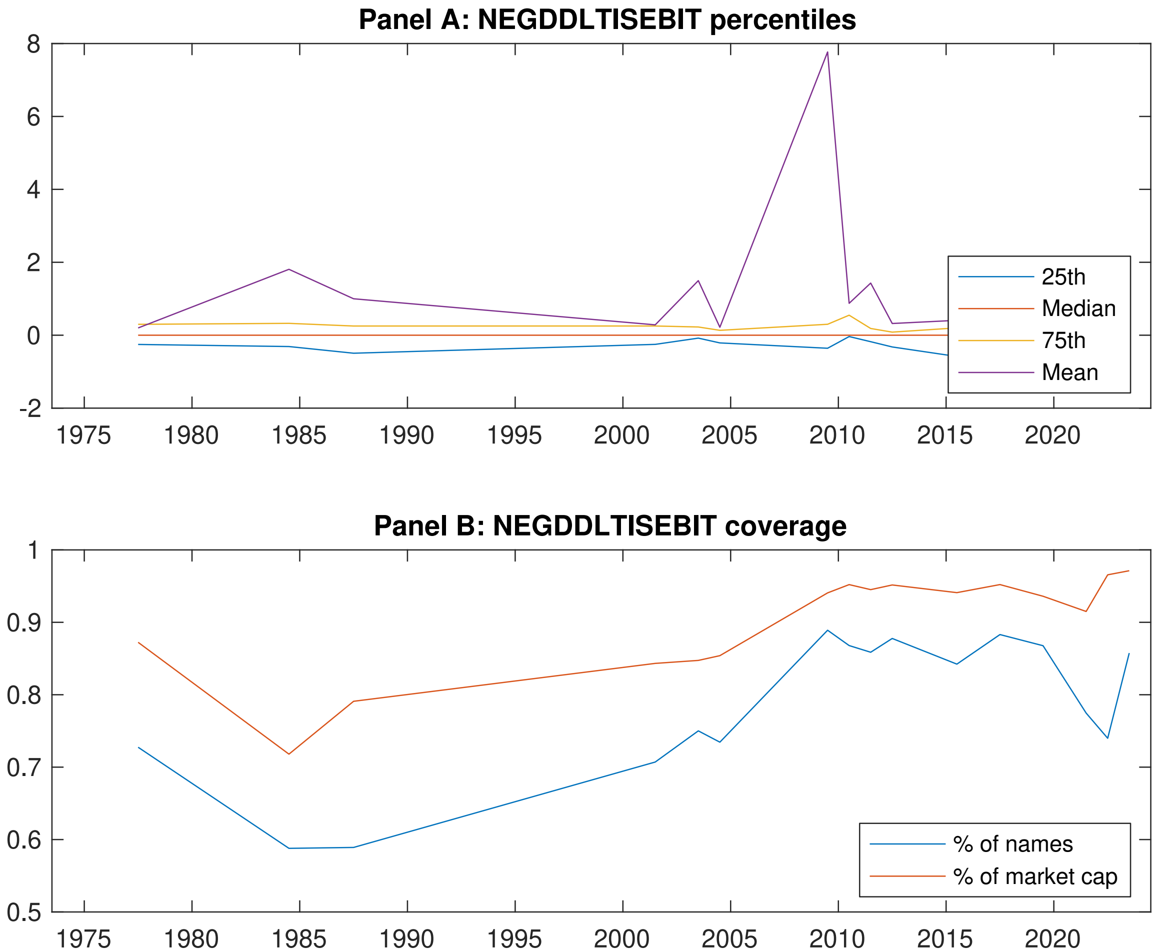


Figure 1: Times series of EDP percentiles and coverage.
This figure plots descriptive statistics for EDP. Panel A shows cross-sectional percentiles of EDP over the sample. Panel B plots the monthly coverage of EDP relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDP. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on EDP-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.56	0.62	0.66	0.78	0.77	0.21
	[2.57]	[3.45]	[3.37]	[4.43]	[3.61]	[2.82]
α_{CAPM}	-0.17	0.01	-0.00	0.18	0.05	0.23
	[-2.84]	[0.25]	[-0.04]	[4.16]	[0.82]	[2.99]
α_{FF3}	-0.22	-0.03	0.05	0.18	0.04	0.26
	[-3.77]	[-0.65]	[1.04]	[4.18]	[0.62]	[3.41]
α_{FF4}	-0.20	-0.00	0.09	0.14	0.03	0.22
	[-3.31]	[-0.10]	[1.78]	[3.36]	[0.45]	[2.93]
α_{FF5}	-0.20	-0.09	0.07	0.11	0.03	0.23
	[-3.44]	[-2.00]	[1.24]	[2.54]	[0.40]	[2.98]
α_{FF6}	-0.19	-0.06	0.10	0.09	0.02	0.21
	[-3.13]	[-1.48]	[1.79]	[2.03]	[0.32]	[2.68]
Panel B: Fama and French (2018) 6-factor model loadings for EDP-sorted portfolios						
β_{MKT}	1.10	0.99	0.99	0.96	1.08	-0.01
	[79.07]	[97.73]	[78.49]	[97.26]	[74.41]	[-0.73]
β_{SMB}	0.16	-0.11	-0.04	-0.04	0.16	-0.00
	[7.85]	[-7.19]	[-1.93]	[-2.55]	[7.30]	[-0.14]
β_{HML}	0.11	0.11	-0.14	-0.04	-0.04	-0.16
	[4.19]	[5.49]	[-5.89]	[-2.11]	[-1.57]	[-4.51]
β_{RMW}	0.03	0.13	0.02	0.08	-0.02	-0.06
	[1.29]	[6.58]	[0.95]	[4.26]	[-0.74]	[-1.59]
β_{CMA}	-0.10	0.06	-0.06	0.15	0.09	0.19
	[-2.37]	[2.08]	[-1.57]	[5.15]	[2.22]	[3.64]
β_{UMD}	-0.03	-0.04	-0.05	0.03	0.01	0.03
	[-1.90]	[-3.61]	[-3.81]	[3.52]	[0.53]	[1.90]
Panel C: Average number of firms (n) and market capitalization (me)						
n	647	527	1060	589	635	
me ($\$10^6$)	1196	3156	2445	3281	1249	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDP strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.21	0.23	0.26	0.22	0.23	0.21
			[2.82]	[2.99]	[3.41]	[2.93]	[2.98]	[2.68]
Quintile	NYSE	EW	0.11	0.11	0.12	0.12	0.12	0.12
			[2.54]	[2.70]	[2.73]	[2.65]	[2.73]	[2.73]
Quintile	Name	VW	0.21	0.24	0.27	0.23	0.21	0.20
			[2.95]	[3.38]	[3.77]	[3.29]	[3.03]	[2.77]
Quintile	Cap	VW	0.21	0.24	0.26	0.22	0.19	0.17
			[3.42]	[3.86]	[4.18]	[3.54]	[3.05]	[2.68]
Decile	NYSE	VW	0.26	0.25	0.25	0.26	0.21	0.22
			[2.60]	[2.52]	[2.50]	[2.49]	[2.04]	[2.10]

Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.15	0.16	0.19	0.17	0.17	0.15
			[2.01]	[2.13]	[2.47]	[2.24]	[2.15]	[1.93]
Quintile	NYSE	EW	-0.15					
			[-2.55]					
Quintile	Name	VW	0.15	0.18	0.20	0.18	0.16	0.15
			[2.09]	[2.55]	[2.85]	[2.61]	[2.29]	[2.08]
Quintile	Cap	VW	0.16	0.19	0.21	0.19	0.16	0.14
			[2.61]	[3.13]	[3.37]	[3.05]	[2.47]	[2.20]
Decile	NYSE	VW	0.18	0.17	0.16	0.17	0.12	0.12
			[1.83]	[1.63]	[1.60]	[1.63]	[1.16]	[1.17]

Table 3: Conditional sort on size and EDP

This table presents results for conditional double sorts on size and EDP. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDP. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDP and short stocks with low EDP .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

		EDP Quintiles					EDP Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.76	0.81	0.90	0.96	0.74	-0.02	-0.01	-0.01	-0.04	0.04	0.01
		[2.82]	[2.87]	[3.34]	[3.25]	[2.73]	[-0.31]	[-0.13]	[-0.08]	[-0.46]	[0.46]	[0.15]
	(2)	0.77	0.84	0.82	0.88	0.87	0.10	0.14	0.11	0.12	0.11	0.11
		[2.91]	[3.34]	[3.21]	[3.50]	[3.40]	[1.34]	[1.73]	[1.43]	[1.46]	[1.39]	[1.40]
	(3)	0.82	0.83	0.84	0.76	0.88	0.05	0.08	0.09	0.08	0.12	0.11
		[3.29]	[3.68]	[3.49]	[3.35]	[3.62]	[0.71]	[0.99]	[1.19]	[1.05]	[1.48]	[1.36]
	(4)	0.67	0.84	0.86	0.76	0.89	0.23	0.25	0.27	0.23	0.30	0.27
		[2.86]	[3.93]	[3.94]	[3.61]	[4.02]	[2.91]	[3.20]	[3.41]	[2.92]	[3.67]	[3.28]
	(5)	0.50	0.60	0.61	0.71	0.73	0.24	0.24	0.28	0.25	0.21	0.20
		[2.51]	[3.33]	[3.12]	[3.90]	[3.75]	[2.75]	[2.81]	[3.22]	[2.89]	[2.39]	[2.25]

Panel B: Portfolio average number of firms and market capitalization

		EDP Quintiles Average n					EDP Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	390	390	391	390	389	37	33	32	33	36
	(2)	106	106	106	107	106	60	61	60	60	61
	(3)	75	75	75	75	75	107	109	104	106	107
	(4)	62	63	63	63	62	226	239	226	237	224
	(5)	58	58	58	58	58	1303	2175	1937	2311	1440

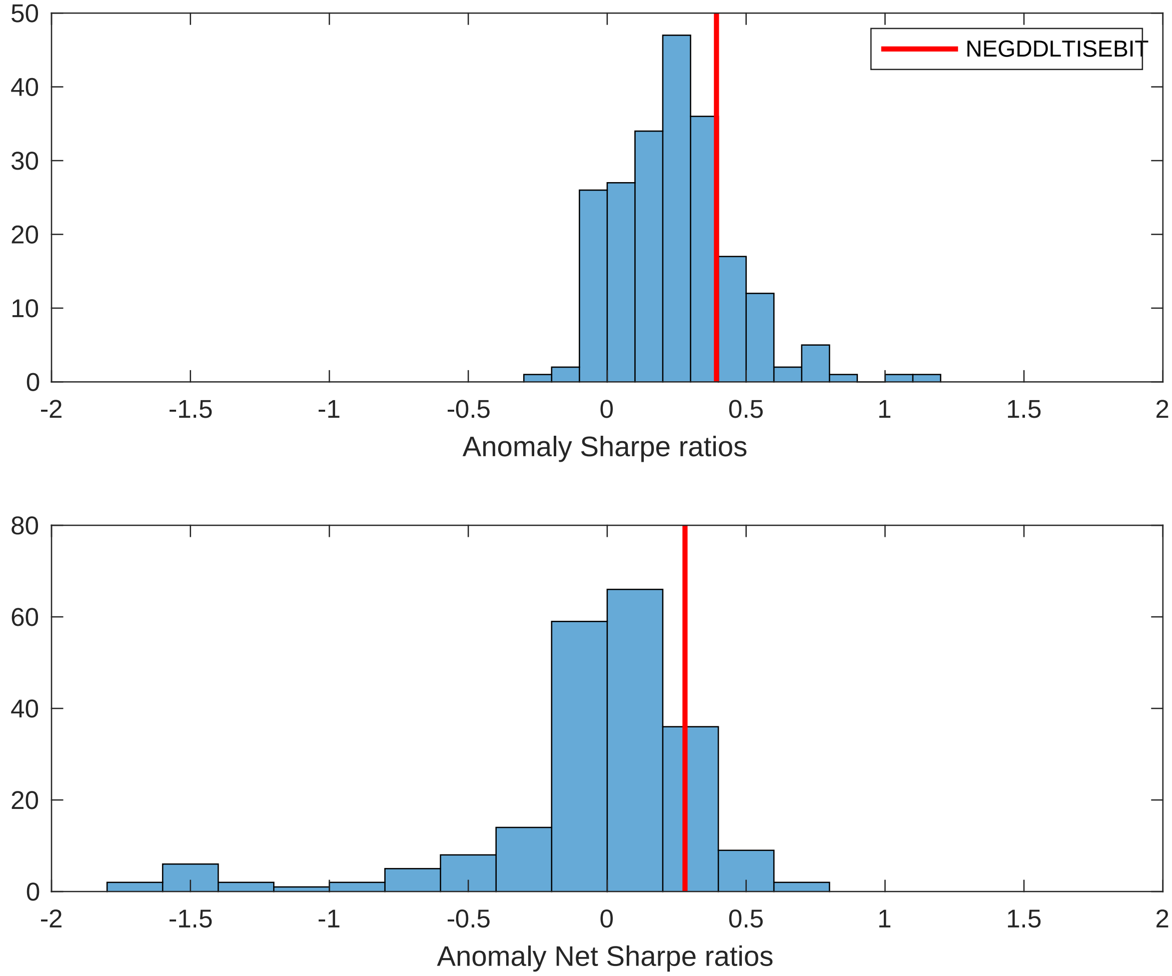


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDP with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

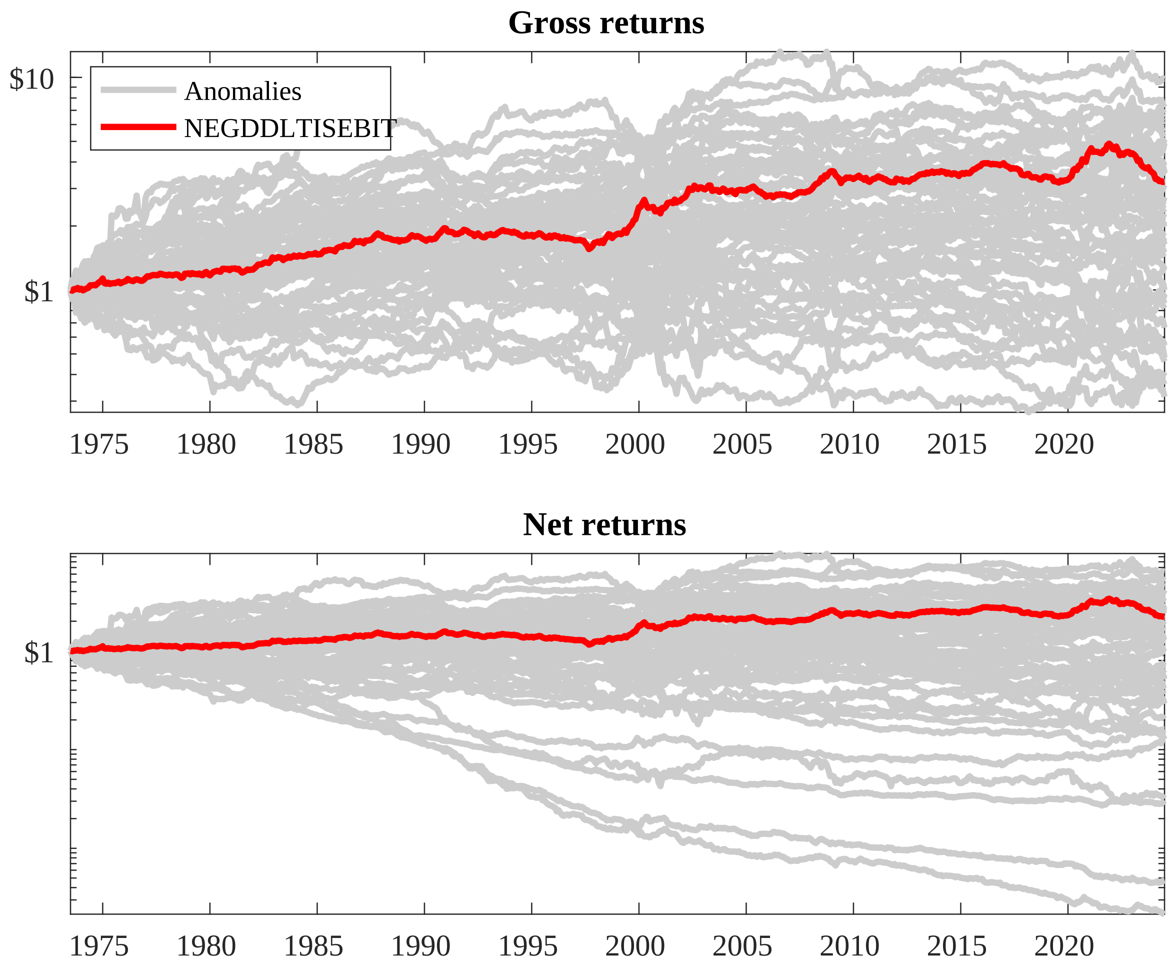
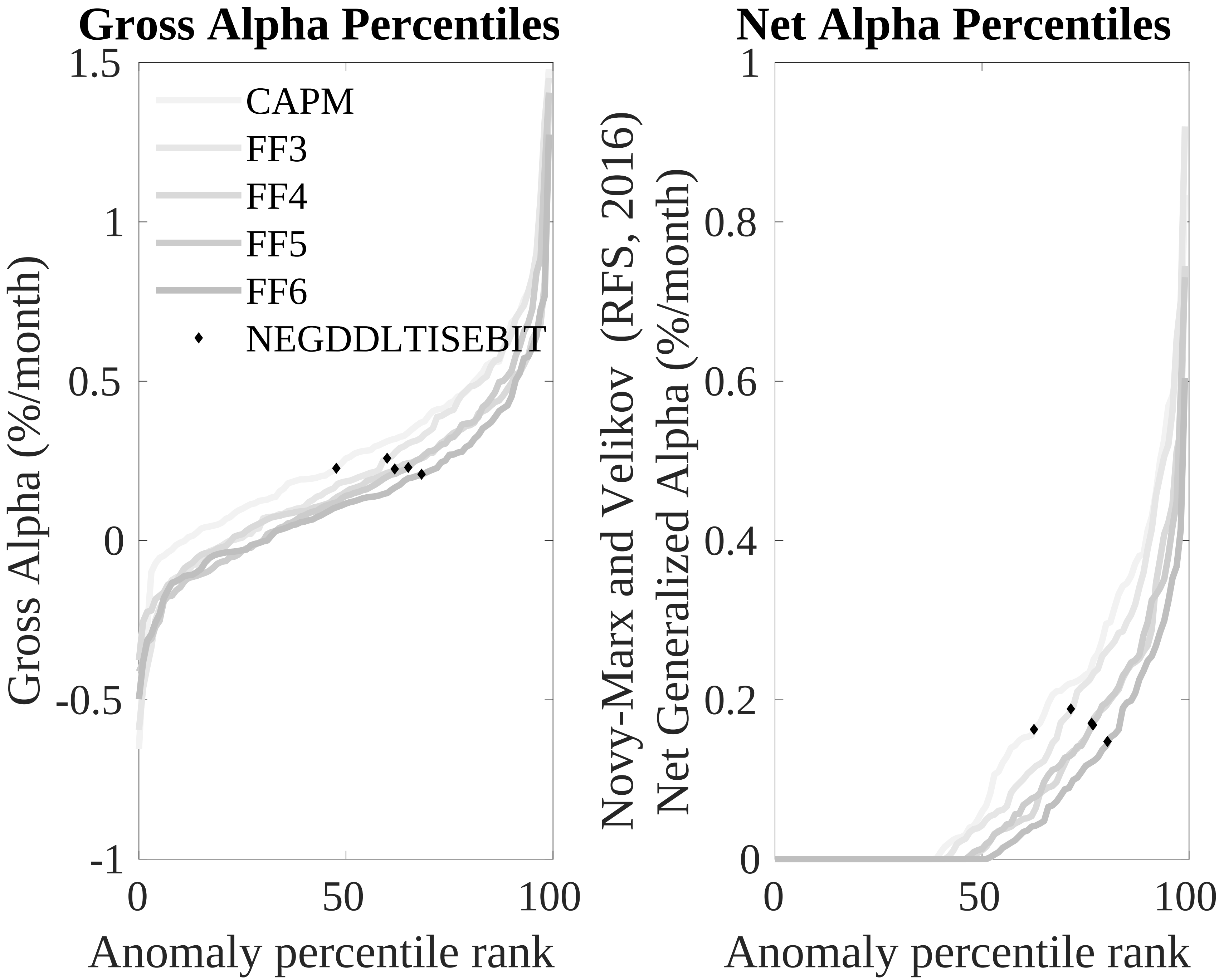
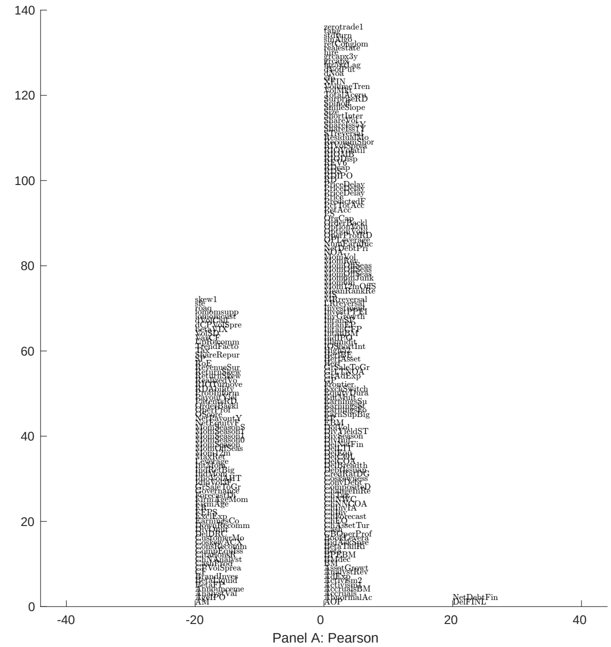


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDP trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDDLTISEBIT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100



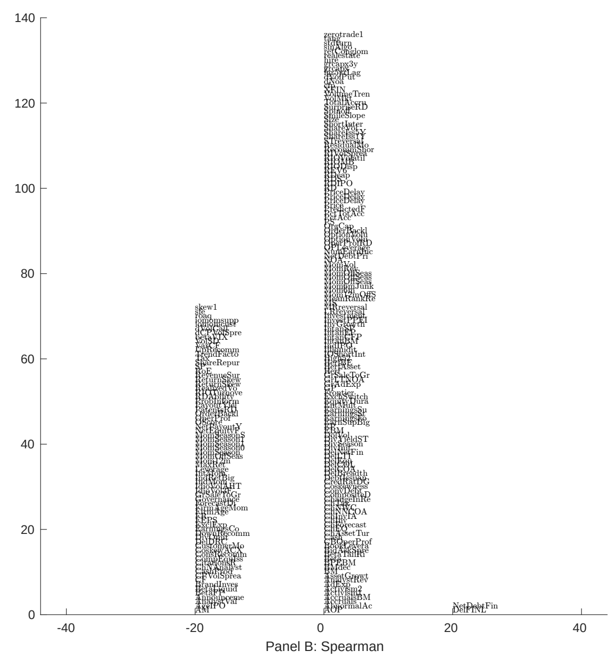


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDP. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

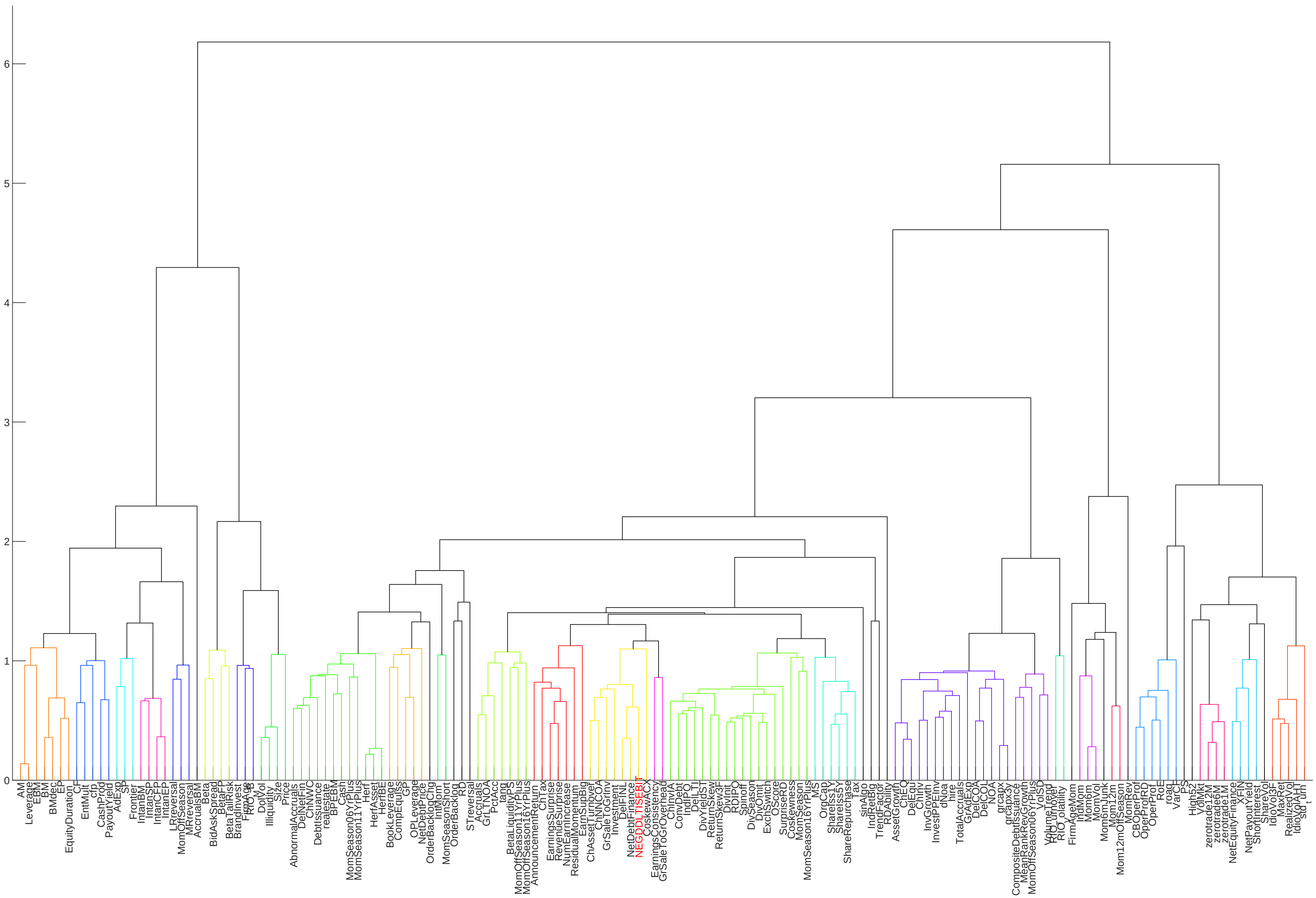


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

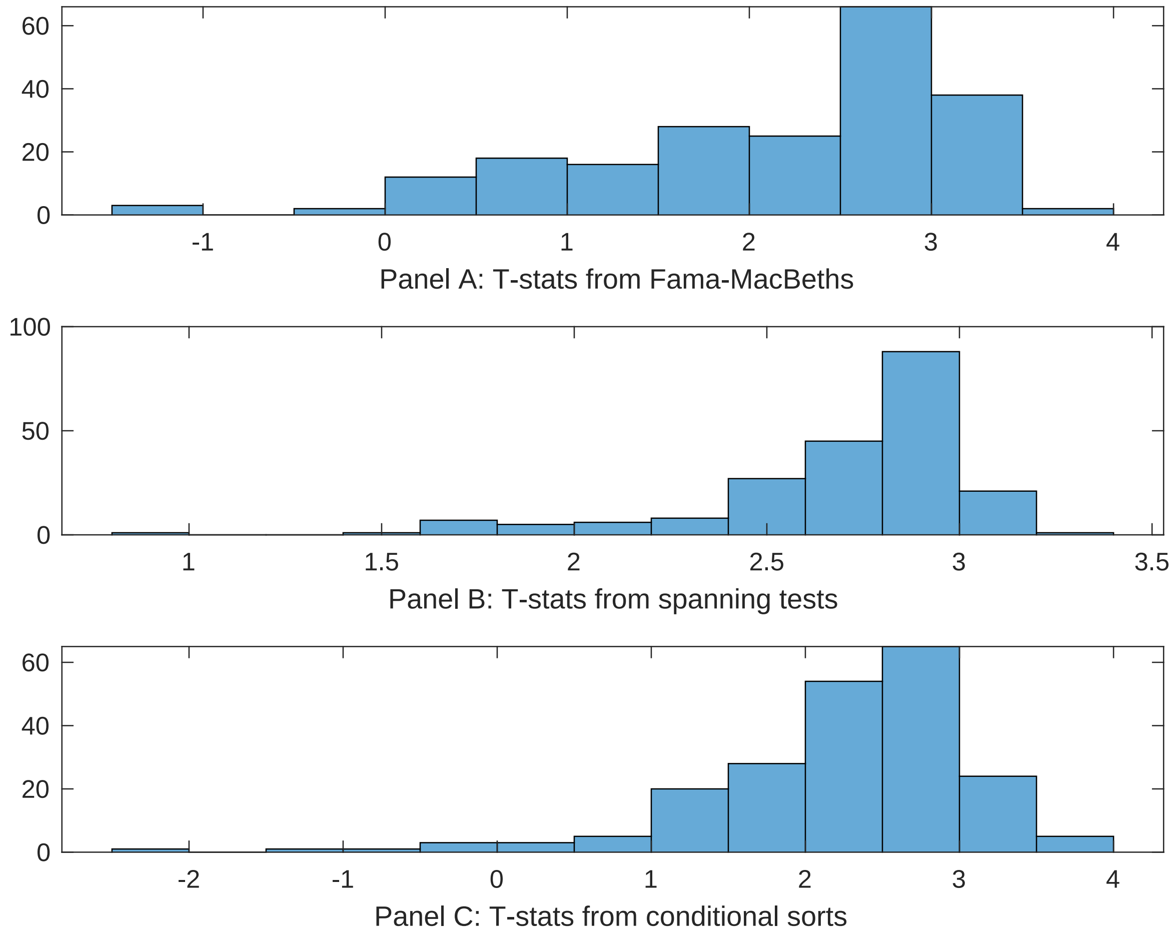


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EDP conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDP} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDP} EDP_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDP,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDP. Stocks are finally grouped into five EDP portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDP trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDP. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDP} EDP_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in financial liabilities, Net debt financing, Change in net financial assets, Investment to revenue, Accruals, Sales growth over inventory growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13	0.13	0.13	0.16	0.12	0.13	0.15
	[5.34]	[5.30]	[5.20]	[6.29]	[4.97]	[5.16]	[6.03]
EDP	0.77	0.20	0.54	0.11	0.92	0.12	0.19
	[0.20]	[0.50]	[1.38]	[2.68]	[2.34]	[2.79]	[0.43]
Anomaly 1	0.18						0.15
	[9.75]						[3.08]
Anomaly 2		0.21					0.85
		[9.43]					[2.03]
Anomaly 3			0.76				-0.85
			[5.14]				[-2.73]
Anomaly 4				0.26			0.20
				[6.05]			[4.28]
Anomaly 5					0.14		0.11
					[4.49]		[2.84]
Anomaly 6						0.16	0.11
						[5.16]	[3.46]
# months	612	612	612	612	612	612	612
$\bar{R}^2$(%)	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the EDP trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDP} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in financial liabilities, Net debt financing, Change in net financial assets, Investment to revenue, Accruals, Sales growth over inventory growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.17	0.18	0.16	0.19	0.18	0.20	0.15
	[2.30]	[2.40]	[2.08]	[2.46]	[2.36]	[2.57]	[1.98]
Anomaly 1	24.62						5.35
	[5.72]						[0.90]
Anomaly 2		24.97					15.02
		[6.02]					[2.79]
Anomaly 3			21.28				11.76
			[5.92]				[2.89]
Anomaly 4				11.15			6.47
				[3.72]			[2.15]
Anomaly 5					6.10		0.55
					[1.98]		[0.18]
Anomaly 6						12.79	7.51
						[3.86]	[2.26]
mkt	-0.59	-1.11	-0.76	-1.40	-0.70	-1.44	-1.01
	[-0.34]	[-0.64]	[-0.43]	[-0.79]	[-0.39]	[-0.81]	[-0.59]
smb	-2.78	-2.46	0.98	-2.32	0.91	-0.79	-2.87
	[-1.04]	[-0.92]	[0.37]	[-0.85]	[0.33]	[-0.29]	[-1.03]
hml	-13.88	-14.50	-15.88	-14.74	-13.94	-15.07	-13.98
	[-4.11]	[-4.31]	[-4.72]	[-4.30]	[-3.95]	[-4.41]	[-4.12]
rmw	-7.85	-8.38	-3.08	-4.69	-3.63	-6.99	-6.43
	[-2.28]	[-2.43]	[-0.90]	[-1.35]	[-0.99]	[-2.00]	[-1.78]
cma	11.42	13.18	25.09	18.60	17.67	18.76	15.72
	[2.15]	[2.53]	[4.83]	[3.57]	[3.31]	[3.61]	[2.82]
umd	1.33	1.86	3.10	2.04	3.01	1.99	0.08
	[0.75]	[1.06]	[1.79]	[1.13]	[1.68]	[1.10]	[0.04]
# months	612	612	612	612	612	612	612
$\bar{R}^2$(%)	10	10	10	7	6	7	14

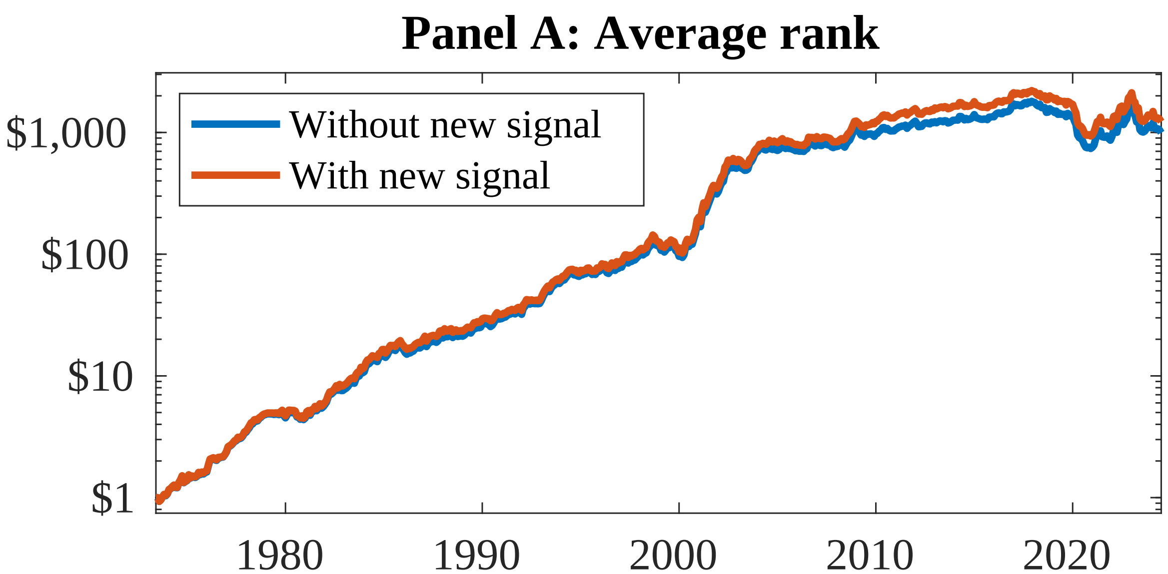


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as EDP. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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